Worksheet 4-5 B Related Rates #2
AP Calculus Name_____

Directions: Solve each problem. You must show all work.

1.) If $x^2 + y^2 = z^2$, find $\frac{dy}{dt}$ when $x = 3, y = 4, \frac{dx}{dt} = -1$, and $\frac{dz}{dt} = 5$.

2.) If $A = \frac{1}{2}bh$, find $\frac{dA}{dt}$ when $b = 7, h = 6, \frac{db}{dt} = 2$, and $\frac{dh}{dt} = -3$.

3.) A particle is moving along the function $y = x^2 + 3$. It is known that $\frac{dx}{dt} = -3$ cm per hour. Find the rate that the value of y is changing with respect to time when $x = 4$.

4.) A 25-foot ladder is leaning against a house. The base of the ladder is pulled away from the house at a rate of 2 feet per second. How fast is the top of the ladder moving down the wall when the base is 7 feet from the house?

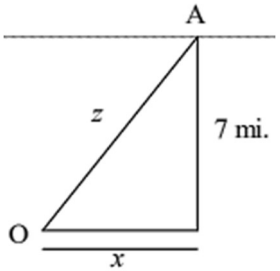
5.) Air is being pumped into a spherical balloon at the rate of 4.5 cubic inches per minute. Find the rate of change of the radius with respect to time when the radius is 2 inches. Volume of a sphere is: $V = \frac{4}{3}\pi r^3$.

6.) The radius of a circle is decreasing at the rate of 6 meters per second. Find the rate that the circumference of the circle is changing with respect to time when the area of the circle is 4π m^2.

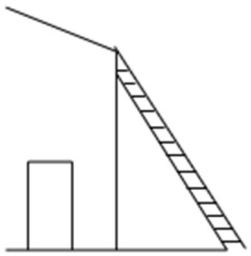
7.) An ice cube is melting at a rate of 5 cubic cm per hour. At what rate is the edge of the cube changing when the edge of the cube is 3 cm.

8.) A circular pool of water is expanding at the rate of 16π in^2/sec. At what rate is the radius expanding when the radius is 4 inches?

9.) An airplane (pt. A) is flying 600 mph on a horizontal path that will take it directly over an observer (pt. O). The airplane maintains a constant altitude of 7 miles (see figure). What is the rate of change of the distance between the observer and the airplane when $x = 5$ miles?



10.) Mr. Brust is using a ladder to paint his house. The 17-ft ladder is leaning against the house when Mr. Kelly decides to pull the base of the ladder away from the house at a rate of 3 ft./sec. How fast is the top of the ladder moving down the side of the house when it is 8 ft. above the ground? Indicate units of measure.



Answers:

1.) $\frac{dy}{dt} = 7$ 2.) $\frac{dA}{dt} = -\frac{9}{2}$ 3.) The value of y is decreasing at a rate of 24 cm/hr

4.) The top of the ladder is moving down the wall at a rate of $\frac{7}{12}$ ft/sec 5.) The radius is increasing at a rate of $\frac{4.5}{16}$ in/min

6.) The circumference is decreasing at a rate of 12π m/sec 7.) The edge of the cube is decreasing at a rate of $\frac{5}{27}$ cm/hr

8.) The radius is expanding at a rate of 2 in/sec

9.) The distance between the observer and the plane is decreasing at a rate of $\frac{3000}{\sqrt{74}}$ mi/hr

10.) The top of the ladder is moving down the side of the house at a rate of $\frac{45}{8}$ ft/sec

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